

❏ 1. Network Authentication: LDAP & NIS

❏ LDAP (Lightweight Directory Access Protocol)

- **Purpose:** Centralized authentication and directory service for users, groups, policies, etc.
- **Structure:** Hierarchical — like a tree (based on DIT - Directory Information Tree).
- **Protocol:** Uses TCP/UDP port 389 (636 for secure LDAP - LDAPS).

❏ Key Components:

Component	Description
slapd	LDAP daemon (OpenLDAP)
ldapsearch	Command-line query tool
ldapadd	Adds entries to the LDAP directory
ldapmodify	Modifies existing LDAP entries
LDIF	LDAP Data Interchange Format

☒ Benefits of LDAP:

- Centralized user and group management
- Integration with services like Samba, Apache, SSH
- Scalable and standardized

☒ NIS (Network Information Service)

- Older centralized authentication system.

- **Developed by** Sun Microsystems.
- Uses port 111 and 834.
- Not secure by modern standards.

☒ NIS Architecture:

Component	Role
Master Server	Maintains database
Slave Server	Redundant copy for failover
Clients	Query the NIS server for user info

☒ LDAP vs NIS:

Feature	LDAP	NIS
Security	Strong (TLS/SSL)	Weak (no encryption)
Scalability	High	Limited
Support	Active	Deprecated
Format	Flexible (custom schemas)	Fixed tables

❏ 2. Apache Clustering

❏ What is Apache Clustering?

A configuration where multiple Apache web servers work together to serve web traffic, usually with shared sessions and coordinated resource management.

❏ Types of Apache Clustering:

1.

Load Balanced Clustering:

- Distribute requests across multiple Apache servers.
- Can use hardware or software load balancers.

2.

Shared File System Clustering:

- Servers access common files (using NFS, GlusterFS).

3.

Session Persistence (Sticky Sessions):

- Ensures user's session sticks to the same server.

4.

Reverse Proxy Clustering:

- Apache acts as a front-facing proxy (e.g., mod_proxy).

☒ **Apache Modules:**

Module	Purpose
mod_proxy	Enables reverse proxy
mod_proxy_balancer	Balancing between backends
mod_ssl	SSL/TLS support
mod_rewrite	URL rewriting

❏ 3. Load Balancer

❏ What is Load Balancing?

Load balancing distributes incoming network traffic across multiple backend servers to ensure:

- High availability
- Redundancy
- Performance optimization

❏ Load Balancer Types:

Type	Description
Layer 4 (Transport)	Based on IP address and TCP/UDP ports

Layer 7 (Application)	Based on HTTP headers, URLs, cookies
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⌘ Load Balancing Algorithms:

Algorithm	Behavior
Round Robin	Sends requests in order
Least Connections	Chooses server with fewest connections
IP Hash	Routes client IP consistently to a server
Weighted Round Robin	Adds weights based on server capacity

⌘ Tools & Technologies:

Tool	Description
HAProxy	High-performance load balancer
NGINX	Web server + load balancer
Apache mod_proxy_balancer	Built-in module for Apache
Keepalived	Provides failover via VRRP
LVS (Linux Virtual Server)	Kernel-level load balancing

⌘ Load Balancer Security:

- Enable HTTPS/SSL termination
- Protect against DDoS (rate limits, IP filtering)

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Use firewalls and WAFs (Web Application Firewalls)